MICROSERVICES ARCHITECTURE, CONTAINERS, AND SERVERLESS COMPUTING: Modern Tech For Digital Transformation

Microservices architecture, containers, and serverless computing are enabling businesses to build more resilient, scalable, and efficient applications. Let's see how these technologies became so popular and the benefits they offer

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Businesses need to embrace a variety of technologies as they transition to the digital age. Perhaps the most important among these is the migration to the cloud. Because the move to the cloud necessitates the creation of more adaptable and independently scalable programs, software engineers are progressively moving away from traditional, monolithic application architectures and towards microservices.

Microservices architecture is a method of software design that divides a large, complicated program into smaller, independent parts (microservices), each of which serves a distinct purpose and may be utilised by other applications. In the past, programmers have created programs with a monolithic design consisting of linked parts that must be changed or scaled simultaneously and executed sequentially. In a microservices software architecture, which divides complicated programs into separate parts, workloads continue moving to the cloud, where scalability is crucial. Containers and serverless software packages that can separate and house individual programs for more effective scalability, maintenance, and transfer between environments, have made this transformation possible.

Fig. 1 depicts the history of modern application deployment over the past two decades, giving a glimpse of the breathtaking pace at which it has evolved.

As we can see in the figure, which shows the evolution of application design and deployment to serverless computing, the adoption of microservices-driven architecture has prevailed. The speed at which services can be booted at significantly low deployment complexity and cost in a pay-per-use model has been a game-changer for businesses. Let us take the example of Netflix and Uber—both industry leaders in their respective domains. They started with small investments in technology that enabled their